

Draft. Public for testnet testing; mainnet not launched.

ENTROPY

Mined Randomness-Utility Token for the GERO Oracle on X1

A mined, hard-capped utility token whose entire non-genesis supply is paid out, line by line, to the node operators and provers of the GERO physical-randomness oracle on X1, by an immutable minter program that nobody holds a key to.

Maximum supply: 21,000,000 ENTROPY

Genesis liquidity: 10% | Mined: 90%

Total allocation: 100%

Version date: October 4, 2026 (v0.1)

Status: draft for technical, economic and legal review. Parameters may be revised before deployment; any revision will be published.

Important Notice

This document describes a utility, participation, emission and liquidity framework for ENTROPY. It is not an offer to sell securities, investment advice, a promise of profit, or a guarantee of token value. ENTROPY is intended to support practical functions inside the GERO randomness oracle and the X1 Forge ecosystem: paying for randomness requests, prepaying the oracle's share of NFT mints, and funding in-game and marketplace mechanics that consume randomness. No return, price appreciation, liquidity level, trading volume or future value is promised.

No third-party audit and no legal review have been performed. The security review of the ENTROPY programs consists of adversarial audit passes run with Claude Code (Fable) in fresh sessions, recorded in the repositories, plus test suites, mutation tests and an end-to-end run against the oracle's line records. No independent audit firm has reviewed the code, and no lawyer has reviewed this document, the token design or the launch. The public site carries the same warning. Section 13 states what was reviewed, how, and what was found and fixed.

The legal characterization of a crypto asset or a transaction involving it depends on facts, circumstances, representations, promises, implementation and applicable law. The U.S. Securities and Exchange Commission's March 2026 interpretation states that non-security crypto assets may nevertheless become subject to an investment contract depending on how they are offered or sold. The design principles in this paper are intended to reduce investment-contract risk through immediate utility, a fixed and program-enforced supply, distribution through protocol participation rather than sale, transparent rules and the avoidance of profit-oriented promises; they do not constitute a legal conclusion.

1. Executive Summary

GERO is a randomness oracle on X1 whose entropy comes from physical radioactive decay measured by Geiger counters run by independent node operators. Every 8 slots it produces one "line": one random result, verified on-chain, that consumers such as Forge Launch use for NFT mints. GERO keeps an on-chain record of every line stating which nodes revealed on time, whether a proof was verified and by whom, and whether the line had demand.

ENTROPY is the token that pays for that work. It is **mined, not pre-minted**: a separate, immutable minter program reads the line record and credits each line's reward to the nodes that were on time, to the prover whose proof was verified, and to an ecosystem vault. Nobody holds a mint key. The supply is hard-capped at **21,000,000 ENTROPY**. **10% (2,100,000) is minted once as genesis liquidity** and paired with 2,000 XNT on Forge Swap, with 60% of the LP tokens burned and 40% locked for 12 months in an immutable lock program. **The remaining 90% (18,900,000) is mined** on a schedule that halves every 16,089,796 lines, about 18 months at the measured X1 slot time. GERO's own upgrade authority, held on a hardware wallet, is the trust root for all emission, because the minter trusts the line records GERO writes.

ENTROPY carries no share of any revenue, pays no yield for holding, and is not staked. Its planned uses are GERO request fees, creator prepayment of the oracle's share on Forge Launch, Shardforge loot and crafting burns, marketplace raffles after a legal review, an ENTROPY/XNT liquidity farm on Forge Farms rewarded in FORGE, and, later, governance of oracle parameters.

2. Design Principles

- **Mined, not pre-minted.** 90% of the supply exists only as it is earned by oracle work. There is no team, investor, treasury or airdrop allocation and no vesting schedule.
- **No mint key.** The mint authority is a PDA of a program that becomes immutable before trading opens; after that the cap, the schedule and the split cannot be changed by anyone.
- **Genesis is liquidity, not an allocation.** All of it is paired with the operator's own XNT in a public pool; 60% of the LP is burned and 40% is locked in an immutable program.
- **Nothing goes unminted except lines with no work.** A line with no on-time node mints nothing; every other line pays out in full; unminted value is never redistributed.
- **Rewards for work, not for holding.** No ENTROPY staking, no fixed APY, no reward for holding or locking ENTROPY or FORGE.
- **Verifiable by anyone, keys disclosed.** Verifiable builds with published hashes, read-only public scripts, a settlement rule anyone can replay from the line record, and every authority, delay and residual power listed in Section 14.

3. Token Specification

Item	Specification
Token	ENTROPY
Network	X1 (Solana-compatible architecture)
Name / symbol	Entropy / ENTROPY (testnet: Entropy (testnet) / tENTROPY)
Token standard	Token-2022

Item	Specification
Extensions / decimals	Exactly two extensions, MetadataPointer and TokenMetadata (no transfer fee, transfer hook, permanent delegate, mint-close authority or confidential transfer); 9 decimals
Maximum supply	21,000,000 ENTROPY (21,000,000,000,000,000 base units), a compile-time constant of the minter
Mint authority	The minter's PDA ["mint_authority"]: 9iNgE4jUDHtWh8cZ36oF2JMaNZbHakRSufGgpvHjqCUn. Nobody holds a mint key
Freeze authority	None
Metadata	Name Entropy, symbol ENTROPY, 9 decimals, description "Entropy (ENTROPY) powers the GERO randomness oracle on X1, used for randomness requests and related on-chain functions. New supply is mined by node operators and provers for their work. Fixed maximum supply 21,000,000. Not an investment; no yield, no revenue or profit share, no expectation of gain." On IPFS or Arweave, URI published at launch; the metadata authorities are held by the deployer until launch and then revoked, with the transactions published
Mainnet mint address	Reserved at zero supply, nothing initialized; addresses published at launch

The mint authority is handed to the minter's PDA before the minter is initialized (irreversible), and the one-shot initialize refuses any mint that does not have exactly this shape with a supply of zero.

4. Supply and Allocation

Maximum supply is fixed at 21,000,000 ENTROPY. Percentages are measured against maximum supply. Genesis is counted inside the cap.

Allocation	%	ENTROPY	Status / purpose
Genesis liquidity	10 %	2,100,000	Minted once by initialize. All of it is paired with 2,000 XNT on Forge Swap in the genesis transaction; 60% of the LP is burned, 40% is locked for 12 months (Section 8)
Mined emission	90 %	18,900,000	Paid per GERO line by the immutable minter on the halving schedule of Section 5. Of every mined line, 80% or 90% goes to on-time nodes, 0% or 10% to the prover, 10% to the ecosystem vault
Total	100%	21,000,000	

Because of integer flooring in each era the minter can never mint the full 18,900,000: the exact maximum is 18,899,999.754037348 ENTROPY (Section 5.2). There is no team, investor, treasury, community or airdrop allocation. Burns never reopen room under the cap, which applies to everything ever minted.

5. Emission Schedule

5.1 Constants

Every constant is compiled into the minter and asserted by its tests; there is no parameter account and no setter (full list in Appendix A). The cap is 21,000,000 ENTROPY, genesis 2,100,000 and emission 18,900,000, all in base units of 10^{-9} . One era is **$N = 16,089,796$ lines** ($\text{round}(547.5 \text{ days} \times 86,400 \text{ s} / (0.3675 \text{ s} \times 8))$), 18 months of 365-day years at the measured slot time). The era-0 reward is **$R_0 = [\text{EMISSION} / (2 \cdot N)] = 587,328,764$ base units = 0.587328764 ENTROPY per line**. $\text{era}(L) = \lfloor (L - L_0) / N \rfloor$, counted in lines from the minter's start line L_0 , not from calendar time, and $R(L) = R_0 \gg \text{era}(L)$. A line is paid only once it is 64 lines (512 slots) old and can no longer change. Cap proof: $\sum_e N \cdot \lfloor R_0 / 2^e \rfloor < 2 \cdot N \cdot R_0 \leq \text{EMISSION}$, checked at compile time and again on every settle and claim.

5.2 Exact totals

Summed over the 30 non-zero eras (0–29) with per-era flooring, total emission is **18,899,999,754,037,348 base units = 18,899,999.754037348 ENTROPY**; genesis plus total emission is 20,999,999,754,037,348, and the shortfall under the cap, **245,962,652 base units = 0.245962652 ENTROPY**, is never minted. Era-0 split per line, in base units: with a verified prover 469,863,011 (node pool) / 58,732,876 (prover) / 58,732,877 (ecosystem); without a prover 528,595,887 / 0 / 58,732,877.

5.3 Era table

Emitted amounts are exact; the cumulative column includes genesis; calendar columns assume 0.3675 s per slot (Section 5.5).

Era	Lines from L_0	Reward per line (ENTROPY)	Emitted in era (ENTROPY)	Cumulative incl. genesis	% of cap	Starts after (estimate)
0	0 – 16,089,796	0.587328764	9,449,999.997692	11,549,999.997692	55.00%	0
1	– 32,179,592	0.293664382	4,724,999.998846	16,274,999.996538	77.50%	≈ 18 months
2	– 48,269,388	0.146832191	2,362,499.999423	18,637,499.995961	88.75%	≈ 36 months
3	– 64,359,184	0.073416095	1,181,249.991667	19,818,749.987628	94.37%	≈ 54 months
...		halving each era				
29	– 482,693,880	0.000000001	0.016089796	20,999,999.754037348	99.999999%	≈ 43.5 years
30 +		0	0			≈ 45 years

5.4 Mined share over time (estimates)

After (at 0.3675 s/slot)	Share of the 18.9 M mined
1.5 years	50%
3 years	75%
4.5 years	87.5%
6 years	93.75%
9 years	98.4%

After (at 0.3675 s/slot)	Share of the 18.9 M mined
15 years	99.9%

Year one is about 17,260 ENTROPY per day and about 6.30 M ENTROPY in total (30.0% of the cap), of which about 630,000 goes to the ecosystem vault. A single node running alone in year one would take about 5.04 M (24.0% of the cap), or about 5.67 M if it were also the only prover. That concentration is why the launch condition is 2–3 independent nodes live before genesis. The starting rate was chosen so that about 87.5% of emission is out after about 4.5 years, after which request fees are intended to carry node operators.

5.5 Slot-time caveat

Halvings count lines, not time. Every calendar figure assumes **0.3675 s per slot**, the X1 mainnet average over the 33 days to 2026-09-27 (0.3728 s since X1's launch in March 2026): one line is 2.94 s, a day about 29,388 lines, an era 547.5 days. If the slot time drifts only the dates move; an era is always 16,089,796 lines (about 14 months at 0.29 s per slot, about 20 at 0.40 s). Halvings will be published as line numbers.

6. How Mining Works

Lines, nodes and provers. GERO divides time into lines of 8 slots. For every line each node commits to a hidden seed ahead of time and reveals it in a short window after the line's bound slot; a reveal inside the window is **on time**. The seeds are combined into the line's result, which a **prover** proves and a separate on-chain verifier checks. A line with no reveal at all is **dead**. Nodes in their 7-day probation (**shadow nodes**) reveal but earn nothing and affect no result. GERO writes one on-chain record per line (a ring of 32,768 lines, about 26.8 hours): which node slots were on time, whether the line had requests, whether a proof was verified and by which prover; its header names each node's payout address. GERO publishes and tests the record invariants the minter relies on. The prover written to a line comes from the verified proof, which is bound to the prover's key, never from whoever sent the transaction.

Who earns what, per line.

Line type	On-time nodes	Prover	Ecosystem vault
Line with a verified prover	80% , split equally among on-time nodes	10%	10%
No-prover line (no verified proof)	90% , split equally	0	10%
Dead line, or any line with zero on-time nodes	0	0	0 (nothing minted)
Shadow node on any line	0	—	—

The ecosystem share takes the rounding dust. The minter never recomputes timing: it reads the on-time bits as GERO recorded them and pays the payout address in the header. Proving is permissionless: the first verified proof recorded wins, and lines without requests normally have no prover and pay 90/0/10.

Settlement and claiming. Anyone can run the permissionless settle crank, which walks the line record in order and credits each payee's claimable balance once a line is 64 lines old and can no longer change. Today the crank is `settle_crank.py`; settlement built into every GERO node's daemon, sending it about every 32 lines, is planned and ships before ENTROPY is initialized on mainnet, with `settle_crank.py` kept as a backup. The Claim page can also send it before claiming. Settlement never stalls, and only lines with zero on-time nodes mint nothing. Each payee opens its own claim account once and can then claim at any time, and tokens are minted only to that payee's own token account; a **Claim page** on the GERO dashboard, linked from X1 Forge, lets a payout wallet do both from a browser.

Immutable minter. The minter's cap, genesis, schedule and split are compile-time constants; it has no admin, pause or parameter, and it becomes immutable before trading opens. If X1 ever drops the program format it was built for, or GERO stops writing its line record, emission halts permanently (Section 16).

7. The Programs

entropy-minter (J79sxnNizAqFpTaXR9C5EAokhYDKW4JkYL4txAq5qAwS, immutable after the launch gate). Holds the mint authority as a PDA and the cap, genesis, schedule and split as constants. Five instructions: `initialize` (one-shot, callable only by the hardware-wallet upgrade authority while the program is still upgradeable; mints genesis and sets the start line), `open_claim`, `settle`, `claim` and `claim_ecosystem`. It checks on every `settle` that the line record is owned by the GERO program at the expected address, and pays the ecosystem share to a token account derived from the vault program's ID, so no key can redirect it. After finalization no key controls it at all.

eco-vault (F7pxRZZcBwHoB19SLRfSe5vSsCPMNDqimu2YMo8ngEHv, upgradeable). Receives the ecosystem share into a token account owned by its own PDA; the minter pins that derivation. An admin key spends under the limits of Section 9, each withdrawal with a permanent on-chain record; a separate upgrade key can veto proposals, reset the admin and upgrade the program with 7 days' notice. It can never inflate supply: it only moves tokens already minted.

lp-timelock (9nmEMo6dtENxgjXgirZzNzMxecxex2tospM1tkF55P3i, immutable before genesis). A minimal generic lock with four instructions: `lock`, `extend` (owner, later only), `burn` (owner) and `withdraw` (beneficiary, after unlock). Nothing can move tokens out early, the unlock time can only move later, and a lock's owner, beneficiary, mint and vault can never change. It has no admin, pause, fee or upgrade path. A lost beneficiary key leaves the tokens locked forever; a lost owner key only removes `extend` and `burn`.

8. Genesis Liquidity

Item	Value
ENTROPY	2,100,000 (10% of cap), minted by <code>initialize</code>
XNT	2,000 XNT from the genesis operator's own funds; not Forge fees, not any Forge vault. The funding transfers are published before genesis
Pool	Forge Swap, Volatile tier (AmmConfig index 1): 1% swap fee, 0.80% to liquidity providers, 0.20% to Forge; 5 XNT pool-creation fee. Not Forge protocol-owned liquidity
Opening ratio	1,050 ENTROPY per XNT (0.000952381 XNT per ENTROPY)
LP	64,807,406,983,978 minted to the creator (100 units stay unminted in the pool): 38,884,444,190,387 burned (60%), 25,922,962,793,591 locked (40%) in <code>lp-timelock</code> until <code>open_time</code> + 365 days; the locked position earns pool fees
<code>open_time</code>	Pool creation + 48 hours; nobody, including the operator, can swap before it

Pool creation, the 60% burn and the 40% lock happen in **one transaction** signed only by the operator's hardware wallet: either the pool exists with 60% burned and 40% locked, or nothing happened. Before signing, the transaction is simulated and the decoded lock's beneficiary, owner, amount and unlock time are asserted against the published values. The minter must be immutable at least 24 hours before `open_time`; if it is not, nothing is announced, the Forge Swap admin disables swaps on the pool before `open_time`, and a new opening is announced at least 24 hours ahead with the lock extended so the 12 months still count from the first trade. Bots at `open_time` are not prevented: the 1% fee and a public start are the only protections. At unlock nothing happens automatically; `relock`, `burn` or `withdraw` is announced at least 30 days ahead, and no option is promised now. Anyone can check the lock with `verify_lock.py` (program hash, lock and vault derivation, amounts, unlock time, roles, and that the locked token is the pool's LP); the burned amount equals the LP supply minus the lock balance minus any later third-party LP.

9. Ecosystem Vault

The vault's only inflow is the minter's permissionless `claim_ecosystem`. In era 0 that is about 1,726 ENTROPY per day and about 630,000 in year one (estimates at 0.3675 s per slot).

Rule	Value
Instant withdrawals	$\leq 10,000$ ENTROPY each and $\leq 25,000$ ENTROPY per rolling 30 days; hard ceilings in code, so parameters can only tighten them
Anything larger	7-day public proposal, executable by anyone after the delay, cancellable by the admin, vetoable by the upgrade authority
Related-party payments (Forge, Shardforge, any project the admin controls or has a stake in)	Always the 7-day proposal, whatever the amount; flagged on-chain; refused on the instant path
Delays and rotation	Limit changes and admin rotation take 7 days; delays are floored at 7 days and capped at 30
Backstop and records	The upgrade authority can veto any proposal and reset the admin at once; every withdrawal leaves a permanent on-chain record (amount, destination, category, purpose, reference hash, related-party flag)

A stolen admin signature can take the instant allowance once and nothing more, provided the upgrade authority acts within 7 days; everything else it can do is public and reversible for 7 days.

First uses: the demand side. Node operators are paid by emission itself. The vault funds projects that use GERO randomness and give ENTROPY a use: integrator grants, developer and tooling grants, Shardforge loot and crafting pools, raffle prize pools (after legal review), fee rebates for projects that pay GERO fees in ENTROPY (draft 20% of fees paid, capped at 5,000 per project per 30 days), audits and bug bounties, and a possible ENTROPY co-reward on the ENTROPY/XNT farm. Every grant has published terms whose hash is the reference on each payout, public decisions, on-chain milestones where possible and payment in tranches; nothing goes to the operator's own wallets or nodes. Node recruitment grants are paid in XNT outside this vault.

Monthly report and upgrades. A read-only script rebuilds the vault report from chain data: every withdrawal with its purpose, reference and transaction, totals, related-party routing, caps, delays, and the balance cross-checked against the minter's counters. Vault upgrades are announced at least 7 days ahead with the diff and the build hash. The admin moves to an independent multisig within about 6 months of launch; no key moves to a multisig program until that program's own upgrade authority is identified or an audited copy is deployed and made immutable.

10. Node Operators

Joining. Nodes are approved by GERO's governance key, held on a hardware wallet. Approval starts a **7-day shadow period** as designed (1,680,000 slots at GERO's 0.36 s per slot assumption; about 7.15 days at the measured 0.3675 s): the node commits and reveals on the real network but affects no result, sets no bits in the line record, is not slashable and earns nothing. After that anyone can activate it. Launch nodes are approved at least 7 days before genesis, and the node set and payout addresses are public from the moment of proposal.

Keys and payouts. Each node has a hot node key in the daemon (commits, reveals, instant self-removal) and a cold owner key never on the node machine. Only the owner key can change the node's **payout address**, and only after a **72-hour public delay** as designed (720,000 slots at GERO's 0.36 s assumption; about 73.5 hours at the measured 0.3675 s). Emission, fees, grants and bond go only to the payout address, so a stolen hot key can cause misses but cannot redirect anything. **On mainnet every node's payout key is a cold, browser-connectable wallet distinct from the hot node key;** it signs `open_claim` once on the Claim page before the minter is initialized. Self-removal is instant; governance removal takes 24 hours; the hot pause key can only pause everything equally for at most 24 hours per use.

Bond and slashing. Nodes post a **bond in XNT, never in ENTROPY**. A node that misses its reveal window is slashable and earns nothing for that line. The current values are a stake of 10 XNT and a slash of 1 XNT per missed line, within governance ranges of stake 1–100 XNT and slash 0.1–10 XNT. Before ENTROPY is initialized, GERO routes slashed XNT to an escrow held by the GERO program that nobody can withdraw from, until GERO v2 defines refunds to affected consumers and a batched ENTROPY buy-and-burn; once any node

other than the Genesis node exists, slashed XNT never goes to the operator. Fuller economics designed in GERO (graduated slashing, an unbonding delay, the refund and buy-and-burn routing) will be published with GERO v2, which ships them.

After the emission tail. About 87.5% of emission is out after about 4.5 years. After that, per-request fees are intended to carry node operators: a fee escrowed on each request and split at line close among the nodes that served it, paid only to payout addresses; a fee payable in ENTROPY, at a discount and with a burn share, is designed for a later version.

11. Utility

ENTROPY complements XNT: consumers can request randomness and mint on Forge Launch in XNT without holding ENTROPY.

Use	Status	Mechanism
GERO request fees in ENTROPY	Designed	A fixed ENTROPY amount per request set by GERO governance with a delay and a cap, at a discount to the XNT fee, with a burn share; no price oracle
Forge Launch: creator prepays the oracle's share	Approved; post-v1 Forge feature	When GERO randomness is used, 5% of the mint price goes to GERO's fee account (Forge's own 5% is always XNT). A creator may prepay that GERO share in ENTROPY at setup, at a per-mint amount read from GERO's on-chain fee parameters; buyers then pay only XNT
Shardforge loot and crafting	Planned	Rolls that use GERO randomness and burn or spend ENTROPY
Marketplace raffles	Designed; launches only after legal review	Tickets in ENTROPY; the randomness request is created only after ticket sales close
ENTROPY/XNT liquidity farm on Forge Farms	Year 1 committed at announcement; years 2-4 planned	40,000,000 FORGE over 4 years as four consecutive 365-day fixed-rate farms: 16 M / 12 M / 8 M / 4 M FORGE (Section 12)
Delegation; oracle-parameter governance	Later, outside the minter	Later delegation stakes XNT: delegated XNT does security work for a node and carries its slashing risk. Never ENTROPY staking for ENTROPY. Governance is not designed yet

Not planned: any ENTROPY stream to FORGE holders or stakers, any ENTROPY reward on the XNT/FORGE farm, and any protocol-level ENTROPY staking yield.

12. Relationship to FORGE

FORGE is the ecosystem token of X1 Forge (Forge Market, Forge Launch, Forge Swap, Forge Farms); ENTROPY is the randomness-utility token of the GERO oracle. They are developed by the same team, and every interaction between them is disclosed and receives no preferential treatment.

Touch point	Rule
Genesis pool	Not Forge protocol-owned liquidity. Its XNT, the 5 XNT pool fee and all rent are the genesis operator's own funds; it pays Forge the same fees as any other pool
Forge Launch and Forge fees	The additional 5% on GERO-enabled mints goes to GERO's fee account, fixed in Forge Launch's code, and is not a Forge fee; Forge's own share stays in XNT. Forge fees received in ENTROPY are treated like any other token. ENTROPY/XNT swaps earn Forge Points like any pair, counting only Forge's protocol share; the genesis operator's wallets and node payout wallets are excluded

Touch point	Rule
Forge Farms	40,000,000 FORGE (4% of FORGE's maximum supply) for the ENTROPY/XNT farm over 4 years: 16 M / 12 M / 8 M / 4 M, year 1 committed, years 2-4 planned, starting at least 7 days after trading opens. Under FORGE White Paper v1.6.3 the Forge Farms allocation is 170,000,000 FORGE with 120,000,000 reserved for future pools; this farm would take 40,000,000 of that reserve. Rewards go only to outside liquidity providers: the burned 60% no longer exists and the locked 40% cannot be staked
FORGE holders	No ENTROPY stream, no share of GERO fees or ENTROPY emission

The FORGE White Paper v1.6.3 does not yet contain these items; the edits are pending for FORGE v1.7.

13. Security and Transparency

13.1 Verifiable builds

All three programs are built with solana-verify 0.5.2 in the image solanafoundation/solana-verifiable-build@sha256:f6b78c98ddb1774af9cf1b5ca00ccff4dac002eded106b4d44560c2c6be4188d (Solana v3.1.14), and every Docker build has been byte-identical to the local build. The program hash is solana-verify get-executable-hash and can be compared with the deployed program using solana-verify get-program-hash.

Program	Build	Program hash
entropy-minter	mainnet	26b53c5473979e1043632b95e382aa1ea72aa7493f4588ed23afaebcdf468502
entropy-minter	testnet (differs only in the compiled-in GERO program ID)	116c96818966757dde52012c43dfbceb0de406c617c9e8e8f479773bc9c5b954
eco-vault		ae80b6f8fc04d482f350fcf3c7b440acd69be859a344bfed491a3700de84716c
lp-timelock		50787f59765db17f3ae590623fa078d60be6ca502cf2f2db78b25f6dc19e837f

Raw file hashes and sizes are in Appendix C. The hashes finalized on mainnet will be the ones re-verified after the last audit pass.

13.2 Audit policy and results

No third-party audit firm is used. Security review means adversarial Claude Code (Fable) audit passes in fresh sessions: attack hypotheses written before reading any source, then source review, proof-of-concept tests against the verified binary, and a review of the existing tests and mutation results, with a written report committed to the repository. Component-level passes are done for the minter, the eco-vault and lp-timelock (two independent passes) and for the oracle version that writes the line record; a **system-level pass** over the whole path is required after the minter is minting on testnet, before launch. **Every finding of any severity is fixed and re-verified with new hashes before launch; none is left open or accepted.**

Program	Hypotheses / PoCs	Findings	Status
entropy-minter	60 / 9	0 Critical, 0 High, 0 Medium, 2 Low, 7 Informational	All fixed
eco-vault	50 / 12	0 Critical, 0 High, 1 Medium, 4 Low, 8 Informational	All fixed
lp-timelock, pass 1	40 / 13	0 Critical, 0 High, 0 Medium, 3 Low, 7 Informational	All fixed (tooling and documentation; program unchanged)
lp-timelock, pass 2	30 / 13	0 Critical, 0 High, 0 Medium, 2 Low, 8 Informational	All fixed (tooling and documentation; program unchanged)

Tests: minter 32 unit and 42 program-level tests, eco-vault 10 and 56, lp-timelock 2 and 53; scripts: crank 13, supply check 16, predeploy 137, eco report 23, eco init 9, verify_lock 29, lock_cli 15; mutation passes of 39/40, 60/60 and 29/30 mutants killed. The verified minter and vault builds were run end to end against the real oracle program on a local validator: 34/34 checks, 53 settled lines paid exactly to the base unit.

13.3 Public verification scripts

All read-only, standard-library Python, RPC reads only:

- `predeploy.py`: the launch gates (deployed hashes, authorities, mint shape, vault derivation, line-record header and ring, launch nodes, Forge Swap configuration, the exact burned and locked LP, the finalized minter, the revoked metadata authorities); `supply_check.py`: minted equals genesis plus claims, accrued plus minted at most the cap, every settled line counted once, mint authority is the PDA, no freeze authority, exact extensions, program final.
- `eco_report.py` (monthly vault report), `settle_crank.py` (permissionless settlement crank), `verify_lock.py` (the genesis lock, lp-timelock repository).

13.4 Testnet lifecycle

On September 30, 2026 the verified testnet build of the minter was initialized on X1 testnet and minted the first ENTROPY: the genesis amount of 2,100,000 tENTROPY, once. The first settlement paid 10 lines, all no-prover lines at 90/0/10.

Check	Result
Per-line split, no-prover line	0.528595887 nodes / 0 prover / 0.058732877 ecosystem vault (90/0/10)
Per-line split, line with a verified prover	0.469863011 nodes / 0.058732876 prover / 0.058732877 ecosystem vault (80/10/10)
Split reconciliation	Both lines reconciled, to the base unit, against the on-chain settlement event of the transaction that paid them
Ecosystem claim	43.168664595 tENTROPY minted to the ecosystem vault
Node claim	403.318661781 tENTROPY minted to the node's payout token account
Supply after both claims	2,100,446.487326376 = genesis 2,100,000 + 403.318661781 + 43.168664595, to the base unit
<code>supply_check.py</code>	Every check passed; the only warnings were the two expected on testnet (metadata authorities not yet revoked, program not yet final)
<code>eco_report.py</code>	Inflow checks passed against the real 43.168664595 inflow

The first ten settled lines were also cross-checked on the oracle side against the line record.

14. Keys and Trust

GERO's upgrade authority is the trust root for all ENTROPY emission. The minter trusts the line record as written by the GERO program. Whoever can upgrade GERO can rewrite the record and redirect the node and prover shares of unsettled and future lines. It cannot raise the per-line reward, touch settled lines, mint for immature lines, exceed the cap or reach the ecosystem share. Today the mainnet GERO upgrade authority is still on a hot key; it moves to the hardware wallet after the cutover to the oracle version with the key-bound verifier and a 3-day soak.

Key	Holder	Power over ENTROPY
GERO upgrade authority	Hardware wallet at launch	Trust root (above)
GERO governance	Hardware wallet	Approve nodes (7-day delay), propose removals (24 h), rotate the pause key; cannot change a node's payout or owner

Key	Holder	Power over ENTROPY
Node owner key / node key / pause key	Cold wallet / hot in the daemon / hot	Change that node's payout after 72 h / commit, reveal, leave; no payout power / pause everything for at most 24 h per use
Minter upgrade authority	Hardware wallet until finalization	The only key that can call initialize and mint genesis; gone after finalization
Ecosystem-vault admin and upgrade authority	Hardware wallet (admin rotatable to a multisig)	Spend under the limits of Section 9 / veto proposals, reset the admin, upgrade with 7 days' notice
lp-timelock owner and beneficiary	Hardware wallet	Extend or burn the locked LP / withdraw it after unlock. The program itself has no authority after finalization
Metadata authorities	Hardware wallet through launch, then revoked	—

Single-seed disclosure. All of these keys are derived from **one hardware wallet and one seed**. Separate keys separate roles against mistakes, not against loss or theft: the seed backup is the single point of failure for every role above. Mitigations: the seed on durable media in two separate places, never photographed or typed into a computer, a tested restore to a second device, and no role key on a hot machine. Until the keys are separated, the vault's upgrade-authority backstop defends against a leaked signature, not a leaked seed. The vault admin, and then its upgrade authority, move to an independent multisig within about 6 months of launch.

15. Regulatory Design Considerations

ENTROPY is designed around protocol participation, oracle functionality, predetermined and program-enforced supply rules, and transparent authorities. Official materials should accurately describe these functions and avoid representations that purchasers should expect profits from the team's future managerial efforts:

- Do not promise or advertise guaranteed returns, fixed APY, passive income, price appreciation or investment profit, and do not describe the genesis LP burn, the GERO fee burn share or any buy-and-burn as mechanisms that guarantee or target a higher ENTROPY price.
- Do not represent ENTROPY holders as owners of GERO, Forge or Echo Hound Labs, or as entitled to any revenue, dividends or profits; node operators and provers are paid for work they perform, and holders receive nothing for holding.
- Do not offer ENTROPY staking yield, fee-revenue sharing, or rewards for holding or locking ENTROPY or FORGE; never pay, distribute or redirect any fee, burn share or slash proceeds to holders for holding, in any token.
- Describe emission as payment for protocol participation and the Forge Farms programme as rewards for providing liquidity, disclosed as a related-party subsidy; keep utility available in XNT without holding ENTROPY; disclose every authority, delay and residual power and that no third-party audit or legal review has been performed; do not list raffles as utility until the legal review is complete; review grant, rebate, farm, raffle and marketing practices under the law applicable when they are implemented.

The SEC's March 17, 2026 interpretation recognizes categories including digital tools and addresses investment-contract analysis, airdrops, protocol mining, protocol staking and wrapping. It also emphasizes that a non-security crypto asset can be offered or sold subject to an investment contract. The Howey test remains binding precedent. SEC staff FAQs of September 25, 2026 state that token buybacks on a functional network, and ongoing work to maintain and improve a functional network, do not by themselves create an investment contract; these are staff views without legal force. Accordingly, this paper adopts conservative terminology and mechanics but does not state that ENTROPY is categorically outside securities laws. **No legal review of this design has been performed.**

16. Risks

- **No third-party audit, no legal review.** A defect missed by every review pass would be permanent in the immutable programs.

- **Immutable-program and interface risk.** The minter and lp-timelock cannot be fixed after finalization. If X1 drops the program format they were built for or changes Token-2022 behaviour they rely on, or if GERO stops writing its line record at the pinned address and format, emission halts permanently and claims may fail with it.
- **Oracle and trust-root risk.** GERO's upgrade authority can redirect unsettled and future emission; the oracle's availability determines whether lines, and therefore emission, exist at all.
- **Concentration in early nodes.** With few nodes each earns a large share of year-one emission (Section 5.4); the 2-3-node launch condition is enforced by choosing the start line, not on-chain.
- **Thin liquidity.** 2,100,000 ENTROPY against 2,000 XNT on the 1% tier; prices may move sharply, bots cannot be stopped at `open_time`, and continuous emission adds sell pressure for years.
- **Related-party farm and integrations.** The ENTROPY/XNT farm is subsidized with FORGE by the same team that develops ENTROPY and GERO; Shardforge and marketplace payments from the vault are related-party payments, all disclosed and routed through the public 7-day path.
- **Key and operational risk.** All critical keys derive from one hardware-wallet seed until the multisig lands; a stolen vault admin signature can take 25,000 ENTROPY at once; if settlement stops (today the settle crank; once the planned daemon settlement ships, every node daemon and every other crank) for longer than the record's ring (about 26.8 hours), lines are lost; a payee that changes its payout address without opening the new claim account loses those shares; calendar estimates depend on X1's slot time.
- **Unmet launch preconditions, regulatory and market risk.** Mainnet GERO is on an earlier version and the line record, hardware-wallet authorities, the slash escrow, cold payout addresses and a second and third node are still ahead; laws and interpretations may change, and raffles may be treated as gambling in many jurisdictions; ENTROPY may lose value, be volatile or illiquid. No value is promised.

17. Launch Sequence

Completed (as of October 4, 2026). All three programs designed, implemented, built verifiably (Section 13.1) and audited at component level with every finding fixed; deployed on X1 testnet at their recorded hashes, the vault initialized, the testnet mint created; the minter initialized and minting on X1 testnet, with both claim paths exercised (Section 13.4); GERO with the line record live on X1 testnet; the real minter run end to end against the line record on a local validator (34/34); launch gate scripts written and tested; on X1 mainnet, the ENTROPY mint reserved at zero supply and the three programs deployed at their recorded hashes, nothing initialized (October 2, 2026).

Remaining, in order.

1. Days of continuous testnet minting (started September 30, 2026; Section 13.4).
2. System-level audit pass; fixes; new hashes; testnet upgrades.
3. Genesis rehearsal on testnet with the hardware wallet, including the push-back path; lp-timelock finalized on testnet; the Claim page exercised against testnet.
4. GERO mainnet path: cutover to the oracle version with the key-bound verifier, a 3-day soak, then the hardware wallet takes the upgrade authorities (on a hot key today); the line record as its own upgrade after it has run on testnet; slashes routed to the program-held escrow (Section 10); a second and third independent node approved at least 7 days before genesis; request fees; cold payout addresses.
5. Forge mainnet (Forge Swap, Launch and Farms live; the GERO fee account wired into Forge Launch; FORGE White Paper v1.7); the Claim page usable before initialize, so the launch nodes' cold payout wallets sign `open_claim` from it.
6. Mainnet genesis (the mint and programs already reserved): lp-timelock finalized after a mainnet smoke test, eco-vault initialized; launch gate → initialize → check → genesis transaction → minter finalized → metadata authorities revoked → final gate (at least 24 h before `open_time`) → announce and publish, with the Claim page public on the GERO dashboard, linked from X1 Forge's navigation, and the site's no-third-party-audit warning; farm year 1 funded; grant terms published.
7. Later: multisig for the vault keys; a possible ENTROPY co-reward on the farm; raffles after legal review; XNT delegation carrying slashing risk (GERO v2); oracle-parameter governance.

Launch gate. Mainnet genesis (initialize, the genesis pool) and finalization happen only after all audit passes are complete, component-level and system-level, and every finding of any severity is fixed and re-verified with new hashes. No program is finalized on mainnet before that gate. The gate does not cover the dashboard or this paper, which are public now so that people can test on X1 testnet.

18. Reference Notes

- U.S. Securities and Exchange Commission, *Application of the Federal Securities Laws to Certain Types of Crypto Assets and Certain Transactions Involving Crypto Assets*, Release Nos. 33-11412 / 34-105020 (Mar. 17, 2026; effective Mar. 23, 2026).
- U.S. Securities and Exchange Commission, *Transactions Involving Crypto Assets* (capital-raising building block; last updated Apr. 29, 2026).
- U.S. Securities and Exchange Commission, *Crypto Assets and the Federal Securities Laws* (capital-raising building block; Apr. 22, 2026, updated May 15, 2026).
- U.S. Securities and Exchange Commission, Division of Corporation Finance, *Frequently Asked Questions: Application of the Federal Securities Laws to Certain Types of Crypto Assets and Certain Transactions Involving Crypto Assets* (Sept. 25, 2026).
- FORGE White Paper v1.6.3 (September 29, 2026), Echo Hound Labs.

19. Glossary

- **Line.** GERO's 8-slot unit of time; each line yields one random result and one ENTROPY payout. **On time:** a node's reveal landing in the window after the line's bound slot. **Dead line:** no reveal at all; mints nothing.
- **Prover.** Whoever submits the line's verified proof; earns 10% of the line. A **no-prover line** has none and pays 90% to nodes, 10% to the ecosystem.
- **Line record.** GERO's on-chain ring of per-line records that the minter reads. **Shadow node:** an approved node in its 7-day probation, earning nothing. **Payout address:** the header address that receives a node's share, changed only by the cold owner key after 72 hours.
- **Era.** 16,089,796 lines; the reward halves at each era boundary, counted from the start line. **Genesis liquidity:** the 2,100,000 ENTROPY minted once and paired with 2,000 XNT.
- **Finalization.** Solana's irreversible removal of a program's upgrade authority. **Verifiable build:** a Docker build in a pinned image whose hash anyone can reproduce and compare with the deployed program.

Appendix A. Constants

CAP = 21_000_000_000_000_000	GENESIS = 2_100_000_000_000_000	EMISSION =
18_900_000_000_000_000		
LINE_SLOTS = 8	LINES_PER_ERA (N) = 16_089_796	R0 =
587_328_764 = floor(EMISSION / (2 * N))		
NODE_BPS = 8_000	PROVER_BPS = 1_000	NODE_BPS_NO_PROVER = 9_000
SETTLE_DELAY = 64 lines (512 slots)		
DECIMALS = 9	Total emission = 18_899_999_754_037_348 (eras 0–29)	Shortfall =
245_962_652		
Era-0 split, prover line: 469_863_011 / 58_732_876 / 58_732_877	no-prover line: 528_595_887 / 0 / 58_732_877	
Genesis LP to creator 64_807_406_983_978; locked (40%) 25_922_962_793_591; burned (60%) 38_884_444_190_387		
open_time = creation + 48 h; unlock_ts = open_time + 31_536_000 s; vault instant limits 10_000 / 25_000 per 30 days; delays 7 days		
Slot time 0.3675 s (33 days to 2026-09-27); line 2.94 s; ≈ 29,388 lines/day; era 547.5 days		

Appendix B. Addresses

Item	Address
entropy-minter program	J79sxxwNizAqFpTaXR9C5EAokhYDKW4JkYL4txAq5qAwS
Mint authority PDA	9iNgE4jUDHtWh8cZ36oF2JMaNZbHakRSufGgpvHjqCUn
eco-vault program	F7pxRZZcBwHoB19SLRfSe5vSsCPMNDqimu2YMo8ngEHv
eco-vault authority PDA	F7UHbWpjRZ7grjgViafUdQWxRG8NXdnA8Kv1HoZ57MQK
lp-timelock program	9nmEMo6dtENxgjXgirZzNzMxecxex2tospM1tkF55P3i
GERO program, mainnet	BxUNG2yo5371BQMZPkfcxdCptFRDHkhvEXNM1QNPBRYU
GERO program, testnet	2dQf9uaCzXewrDNLttmtzQmc3SmqfAHz3qahKQjtGQyY
tENTROPY mint, testnet	85xvCxxKSns83kbwjyqkAuSgVyAd3ZDdYfgVEfTCgTnm
ENTROPY mint, pool, LP mint, lock: mainnet	Reserved at zero supply, nothing initialized; addresses published at launch

Appendix C. Raw build hashes

Program hashes are in Section 13.1. Raw file SHA-256 and sizes: entropy-minter mainnet 8207a91e5200297a69c013db61cbcd5429a169add72a869714eccc59f0669075 (387,296 B); entropy-minter testnet e2b341f40171836f9fa2a9ced791648848b2a0d21dcc58300ca38fe62949d927 (387,296 B); eco-vault 498fbace8b6d3068862e0b4a395f23b7fca3ccf5a4a0687b0b0bb2c55b166846 (512,544 B); lp-timelock ea811bebafe1f13912831a79755e7fe20b7ce3c785038ced428667fcc8b6420 (279,912 B). Build image solanafoundation/solana-verifiable-build@sha256:f6b78c98ddb1774af9cf1b5ca00ccff4dac002eded106b4d44560c2c6be4188d.

Status: Draft. Public for testnet testing; mainnet not launched. Nothing in this document guarantees token value, liquidity, rewards, trading volume or regulatory treatment. No third-party audit and no legal review have been performed.